

FIND-S Algorithm

To upload file

```
from google.colab import files
```

```
uploaded = files.upload()
```

To read the uploaded file

```
import pandas as pd
```

```
data = pd.read_csv("play_tennis_dataset.csv")
```

```
data=data.head(5)
```

```
print(data)
```

Concepts and Target

```
concepts = data.iloc[:, 1:-1].values
```

```
target = data.iloc[:, -1].values
```

```
print("Features:\n", concepts)
```

```
print("\nTarget:\n", target)
```

```
... Features:
  [['Overcast' 'Mild' 'Normal' 'Strong']
   ['Sunny' 'Mild' 'Normal' 'Strong']
   [nan 'Mild' 'High' 'Strong']
   ['Sunny' 'Mild' 'High' 'Weak']
   ['Sunny' 'Cool' 'Normal' 'Strong']]

Target:
  ['Yes' 'Yes' 'No' 'Yes' 'Yes']
```

To Find-S Algorithm

```
hypothesis = None
```

```
for i in range(len(target)):
```

```
    if target[i] == "Yes":
```

```
        if hypothesis is None:
```

```

        hypothesis = concepts[i].copy()
    else:
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = "s"

    print(f"Hypothesis after positive example {i+1}:")
    print(hypothesis)

print("\nFinal Hypothesis:")
print(hypothesis)

```

Result:

```

... Hypothesis after positive example 1:
    ['Overcast' 'Mild' 'Normal' 'Strong']
Hypothesis after positive example 2:
    ['s' 'Mild' 'Normal' 'Strong']
Hypothesis after positive example 4:
    ['s' 'Mild' 's' 's']
Hypothesis after positive example 5:
    ['s' 's' 's' 's']

Final Hypothesis:
    ['s' 's' 's' 's']

```